Nami Naziri

Espoo, Finland

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Personal Profile

Game programmer passionate about gameplay, animation, and AI systems. Experienced in Unreal Engine with a strong foundation in C++. I enjoy collaborating on challenging systems, solving complex problems, and learning new technologies. I'm excited to contribute to ambitious teams that push the boundaries of interactive experiences.

Skills

Programming C++, C, Python, JavaScript
Software **Commercial Engine** (Unreal Engine), **Source Control** (Git, Perforce, Plastic SCM), Docker, Maya
Animation Animation Blueprint, Motion Matching, State Machines, Blend Spaces, Pose warping, Inverse kinematics
AI Behavior tree, State Tree, Mass AI
Soft Skills Communication, Teamwork, Problem-Solving, Adaptability, Initiative

Work Experience

Programmer, Character Locomotion Development (Motion Matching)

ENDS | Concrete Realm | **STEAM** *Jan 2025 - Present*

- Collaborated with a small, **agile team** to design and implement **locomotion** and **combat systems** in Unreal Engine 5, utilizing **motion matching** with **foot locking**, **overlay systems**, and **stride and orientation warping** to enhance animation quality.
- Collaborated with designers to refine character movement and combat mechanics, optimizing the **3Cs** (Character, Control, Camera) for fluid, responsive locomotion, traversal, and combat interactions.
- Designed and implemented the **hand-to-hand combat system** with combo mechanics and paired animations, ensuring impactful, fluid encounters with responsive and natural character interactions.
- Created the **interaction system** for dynamic object interactions, utilizing **motion warping**, hand **IK**, and **motion matching** to ensure contextually appropriate animations and precise character-object alignment
- Integrated the **Mutable plugin** to develop a **character customization system**, including UI implementation and interaction mechanisms.
- Designed and implemented camera systems for locomotion, combat, and character customization using the **Gameplay Camera plugin**.

Research Assistant - Animation & Machine Learning

Aalto University *April 2024 - June 2024*

- Developed advanced character animation systems using state-of-the-art machine learning technologies, including **VAEs** and **GANs**.
- Worked on both kinematic and physics-based animation generation, creating realistic and adaptable character behaviors.

Freelance AI Programmer

Richard Rendering *April 2023 - March 2024*

- Designed and implemented **NPC behaviors** using **behavior trees** and **state trees**, and integrated **smart objects** for dynamic interactions (e.g., mantling with Nav Link Proxies, companion AI, and contextual interactions).
- Developed the **crowd system** using C++ and the **multithreaded MassEntity** framework, enhancing it with **accessory and animation** systems for agents, and addressing performance bottlenecks, resulting in significant optimization and improved frame rates by a factor of two.
- **Collaborated with cross-functional teams** to ensure seamless integration of systems with core game features.

Education

Aalto University *Espoo, Finland*

Master's Degree, Computer Science (GPA: 4.64/5) *2022 - 2024*

- **Thesis** Controllable Part-wise Motion Generation: A machine learning approach for generating natural character animations by separating upper and lower body movements while preserving harmony between them.
- **Scholarship** Recipient of the Finland Scholarship, a highly competitive merit-based award covering 100% of tuition fees and €5000 for first-year relocation and living expenses.

Projects

Horse Locomotion System

- Designed and implemented a high-level animation system for horse locomotion using **motion matching** in Unreal Engine 5, achieving smooth transitions between movement states despite a limited animation dataset.
- Implemented **player-horse interactions** (mounting/dismounting) with emphasis on animation quality and realism, ensuring proper alignment and natural motion by warping the motion to match character positioning and movement constraints.
- Applied **inverse kinematics** for realistic hand and reins interactions, enhancing player immersion.
- Trailer: **YouTube** | Download: **Google Drive**

Lost Radiance

- Developed Lost Radiance in Unreal Engine 5 for the **Games Now! Game Jam 2023**: A cosmic survival game where players embody a star battling shadowy entities in a desolate universe.
- Selected for exhibition at **Kumma Gallery** by industry judges (**Supercell, Housemarque, Futureplay, Platonic Partnership**), praised for engaging mechanics and strong visual presentation.
- Trailer: **YouTube** | Download: **itch.io**